



Project Report -1

Wazuh SIEM
Deployment



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Objective : Setting up a Production-like wazuh
SIEM environment for SOC practice.

Date : 04/10/2026

01 Environment

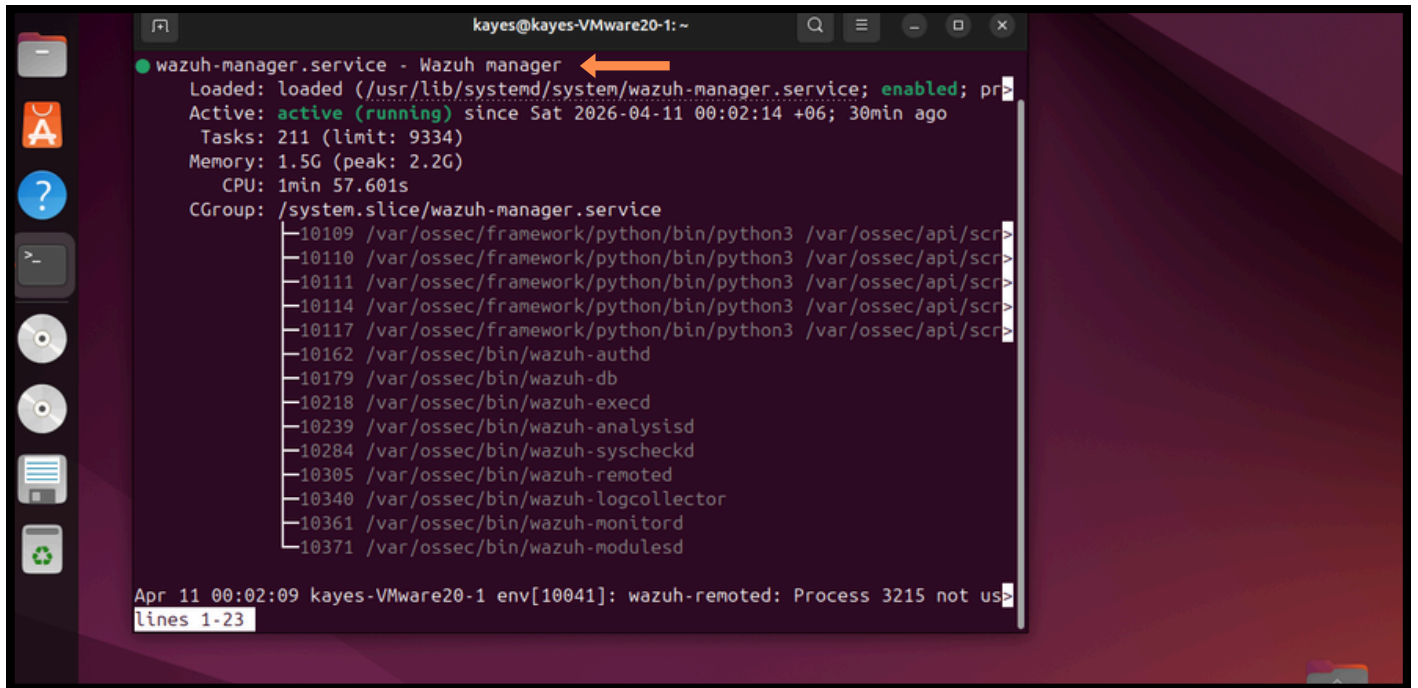
- Host : VMware Workstation
- OS : Ubuntu 24.04 LTS
- Wazuh-Version: 4.14 All-in-One

02. STEP-BY-STEP DEPLOYMENT

1. Ran the official one-line installer :
“curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh && sudo bash wazuh-install.sh -a”
2. Noted Admin Password from “wazuh-password.txt”
3. Accessed dashboard at “https://192.168.0.106”



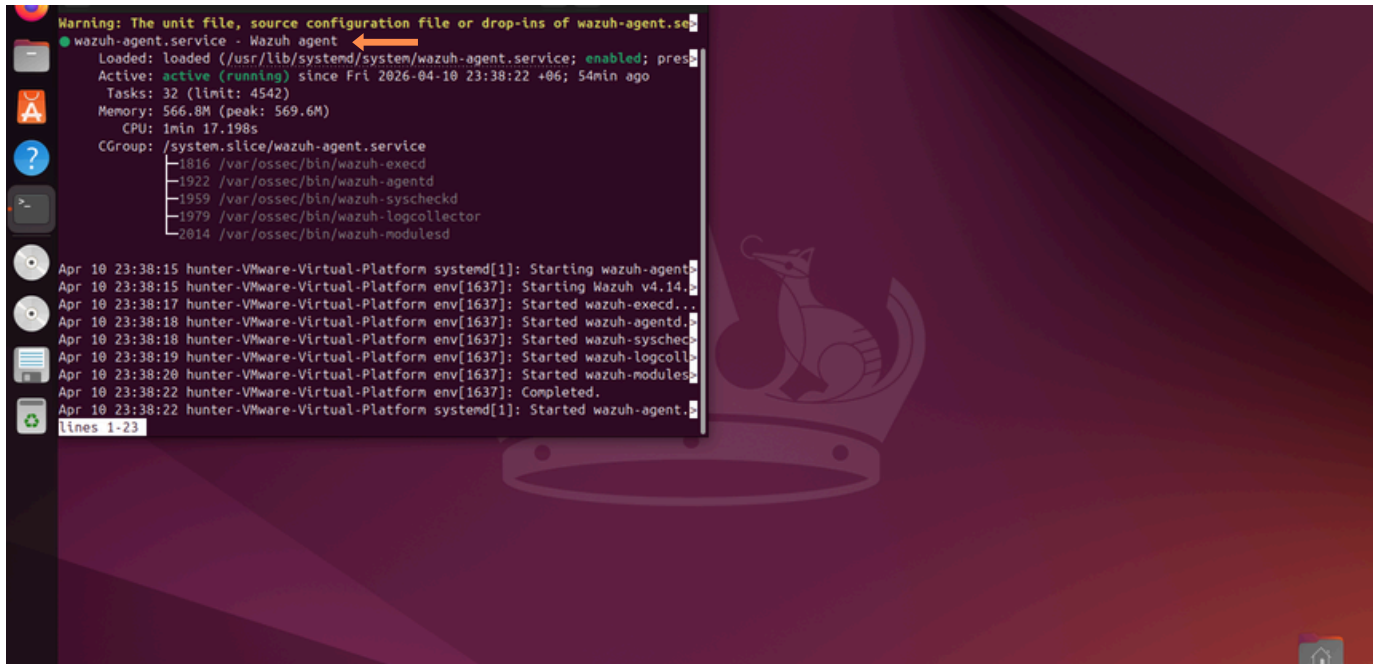
03 Screenshots



```
kayes@kayes-VMware20-1: ~  
● wazuh-manager.service - Wazuh manager  
Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; pr>  
Active: active (running) since Sat 2026-04-11 00:02:14 +06; 30min ago  
Tasks: 211 (limit: 9334)  
Memory: 1.5G (peak: 2.2G)  
CPU: 1min 57.601s  
CGroup: /system.slice/wazuh-manager.service  
├─10109 /var/ossec/framework/python/bin/python3 /var/ossec/api/scr>  
├─10110 /var/ossec/framework/python/bin/python3 /var/ossec/api/scr>  
├─10111 /var/ossec/framework/python/bin/python3 /var/ossec/api/scr>  
├─10114 /var/ossec/framework/python/bin/python3 /var/ossec/api/scr>  
├─10117 /var/ossec/framework/python/bin/python3 /var/ossec/api/scr>  
├─10162 /var/ossec/bin/wazuh-authd  
├─10179 /var/ossec/bin/wazuh-db  
├─10218 /var/ossec/bin/wazuh-execd  
├─10239 /var/ossec/bin/wazuh-analysisd  
├─10284 /var/ossec/bin/wazuh-syscheckd  
├─10305 /var/ossec/bin/wazuh-remoted  
├─10340 /var/ossec/bin/wazuh-logcollector  
├─10361 /var/ossec/bin/wazuh-monitord  
└─10371 /var/ossec/bin/wazuh-modulesd  
  
Apr 11 00:02:09 kayes-VMware20-1 env[10041]: wazuh-remoted: Process 3215 not us>  
lines 1-23
```

Screenshot 01 : Wazuh manager installed and active. To do this i had to run commands like “sudo systemctl start wazuh-manager” and “ sudo systemctl status wazuh-manager”.

Screenshots

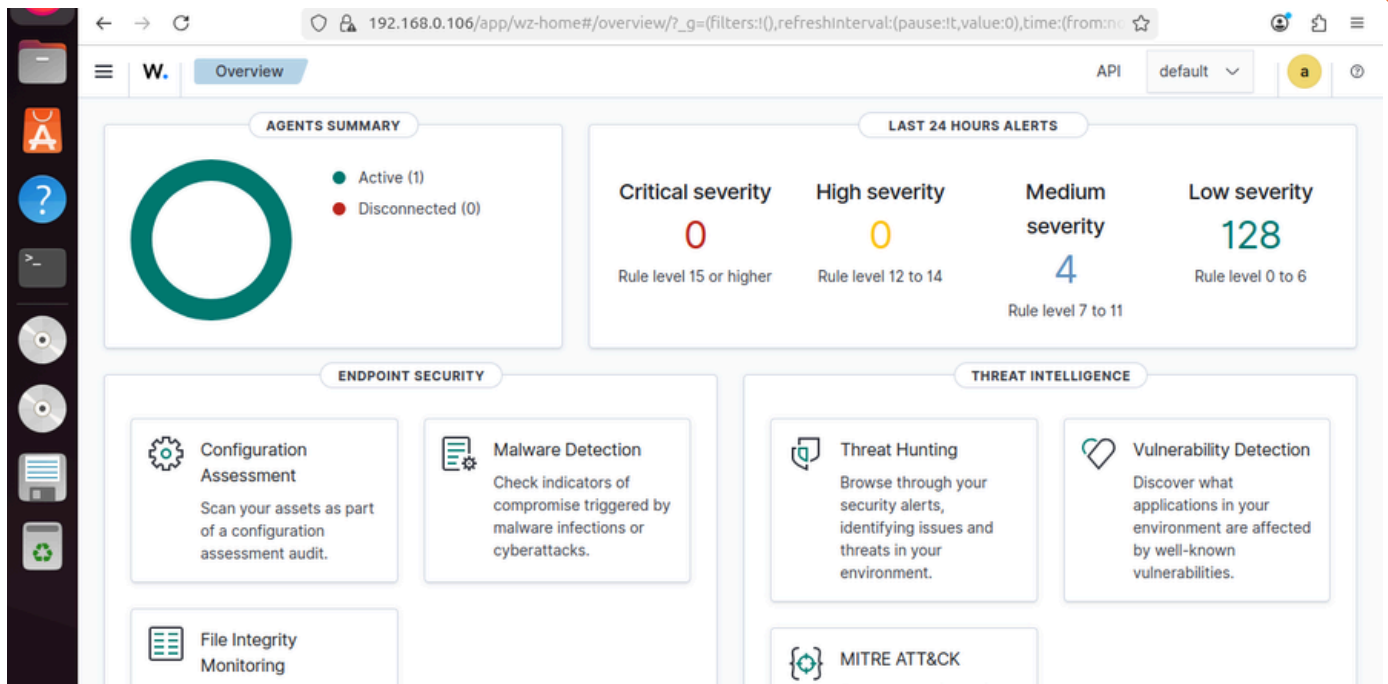


```
Warning: The unit file, source configuration file or drop-ins of wazuh-agent.se
● wazuh-agent.service - Wazuh agent
   Loaded: loaded (/usr/lib/systemd/system/wazuh-agent.service; enabled; pres
   Active: active (running) since Fri 2026-04-10 23:38:22 +06; 54min ago
     Tasks: 32 (limit: 4542)
    Memory: 566.8M (peak: 569.6M)
       CPU: 1min 17.198s
    CGroup: /system.slice/wazuh-agent.service
            └─1816 /var/ossec/bin/wazuh-execd
              1922 /var/ossec/bin/wazuh-agentd
              1959 /var/ossec/bin/wazuh-syscheckd
              1979 /var/ossec/bin/wazuh-logcollector
              2014 /var/ossec/bin/wazuh-modulesd

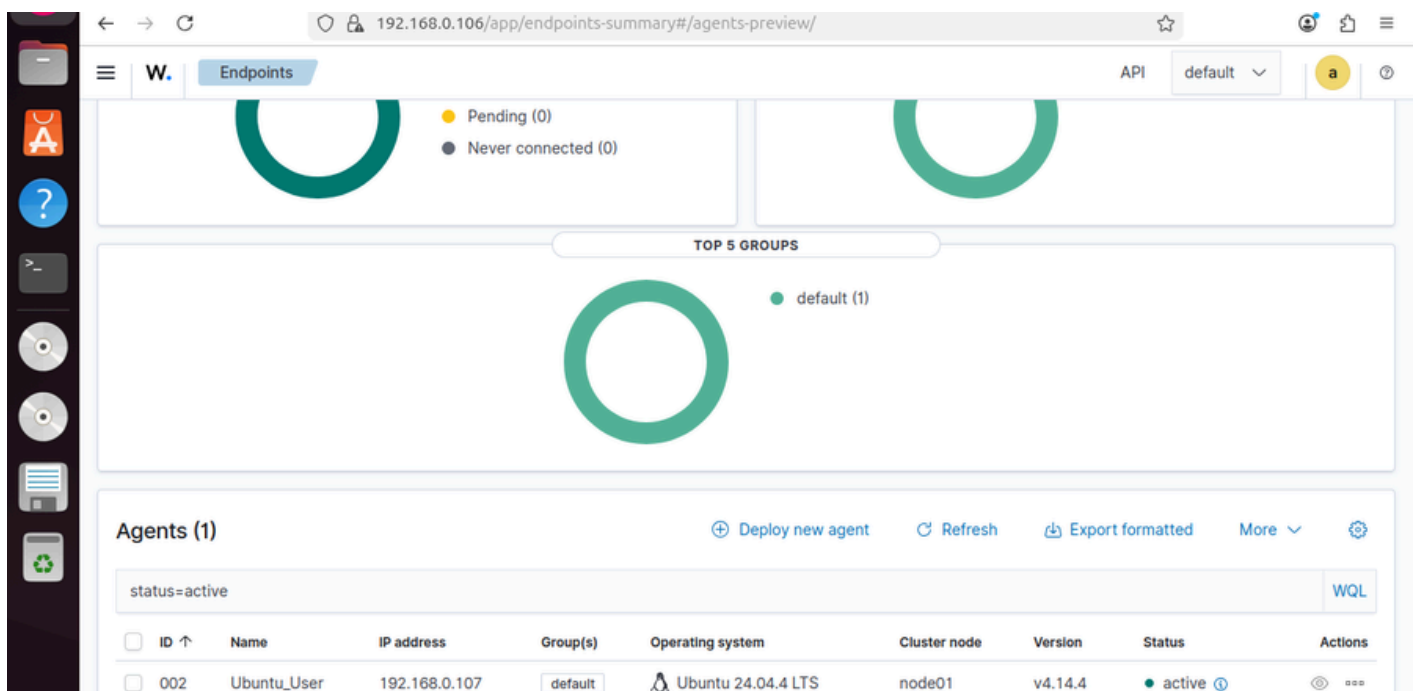
Apr 10 23:38:15 hunter-VMware-Virtual-Platform systemd[1]: Starting wazuh-agent
Apr 10 23:38:15 hunter-VMware-Virtual-Platform env[1637]: Starting Wazuh v4.14.
Apr 10 23:38:17 hunter-VMware-Virtual-Platform env[1637]: Started wazuh-execd...
Apr 10 23:38:18 hunter-VMware-Virtual-Platform env[1637]: Started wazuh-agentd...
Apr 10 23:38:18 hunter-VMware-Virtual-Platform env[1637]: Started wazuh-syscheckd...
Apr 10 23:38:19 hunter-VMware-Virtual-Platform env[1637]: Started wazuh-logcoll...
Apr 10 23:38:20 hunter-VMware-Virtual-Platform env[1637]: Started wazuh-modulesd...
Apr 10 23:38:22 hunter-VMware-Virtual-Platform env[1637]: Completed.
Apr 10 23:38:22 hunter-VMware-Virtual-Platform systemd[1]: Started wazuh-agent.
```

Screenshot 02 : Wazuh Agent Ubuntu_User is active and running. After I have copied the Agent connection one line command “`wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.4-1_amd64.deb && \sudo WAZUH_MANAGER='192.168.0.107'WAZUH_AGENT_NAME='Ubuntu_User' dpkg -i ./wazuh-agent_4.14.4-1_amd64.deb`” to the Ubuntu_User machine terminal and connected it. To check the agent is connected or not I ran the command “`sudo systemctl status wazuh-agent`” in the Ubuntu_User machine’s Terminal window.

Screenshots

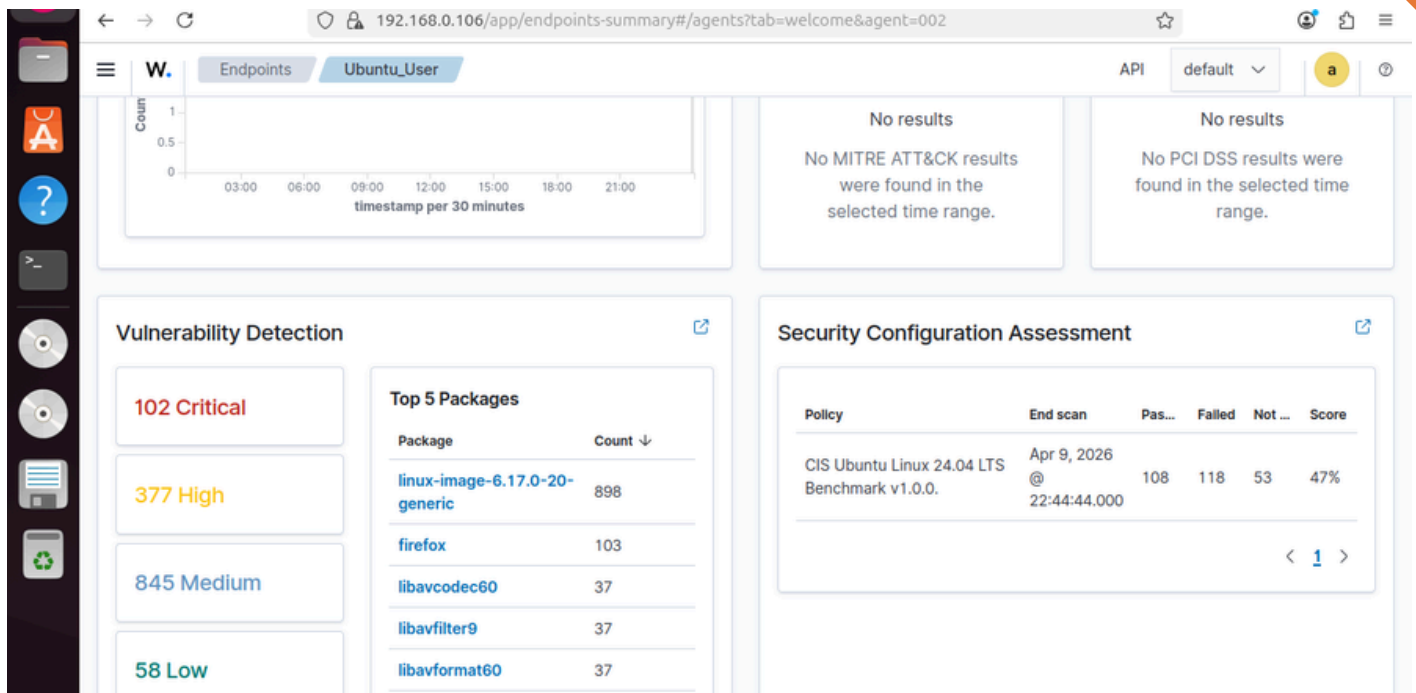


Screenshot 03 : Wazuh Dashboard admin login was Successful. Its showing vulnerability alerts of the wazuh-server host machine . As we can see in the screenshot wazuh is showing 128 low severity alerts and 4 mid severity alerts.



Screenshot 04 : Wazuh Agent tab is showing Ubuntu_User agent is active and its OS info which is Ubuntu 24.04.4LTS and IP-Address 192.168.0.107. As you can see the status is showing Active meaning the Agent is connected and sending log data.

Screenshots



Screenshot 05 : Within the Ubuntu_User agent tab, we can see Vulnerability Detection Alerts of that System. As we can see on the screenshot wazuh has detected 102 Critical alerts , 377 High alerts, 845 Medium alerts , 58 Low alerts accurately.

04 Challenges & Solutions

-Challenge : Self-signed certificate warning → Solution: “Had to manually install curl certifications first then run the wazuh installation command for a successful install.”

-Challenge: Firewall blocking port 5601 → Solution: “Checked the browser option to allow anyway”

-Challenge : Ran into some config error when setting up the Agent with the host server
→Solution : “Had to manually check both config files and correct the Gateway path to same IP for the connection to be Successful.”

05. SOC Skills Gained

- Understanding of SIEM architecture (Manager + Indexer + Dashboard)
- Agent deployment and communication
- Initial alert triage readiness
- Documentation & reporting